

Industry Article

Understanding Gerber Layers for PCBs and Stencils

January 25, 2024 by [Carmen Zheng, NextPCB](#)

Gerber layers don't have to be a mystery. Knowing the names and purposes of each Gerber layer used in PCB manufacturing can improve your designs and ease communication with the board house.

This is the second installment in a three-part article series in which we will:

1. [Explain the Gerber file format](#), file names, and standards revisions.
2. Help you understand the Gerber layers used for PCB and stencil manufacturing.
3. Explain the importance of [Gerber Design for Manufacturing \(DFM\) checks](#) and introduce free resources.

Through this series, we will demonstrate why it is important to check your Gerber file before sending it to the PCB manufacturer, help you identify and avoid common problems, and describe Gerber file checks that you should be performing.

If you read part 1 of this article series, you have a [background on the Gerber format and file names](#). This article will provide a layer-by-layer explanation of the name, abbreviation, and purpose for each layer in a basic PCB.

Gerber Layer Orientation

Gerber files are viewed as if you were looking through the PCB from the top to the bottom so the vias, cut-outs, and traces line up. That means the bottom layers will all appear reversed.

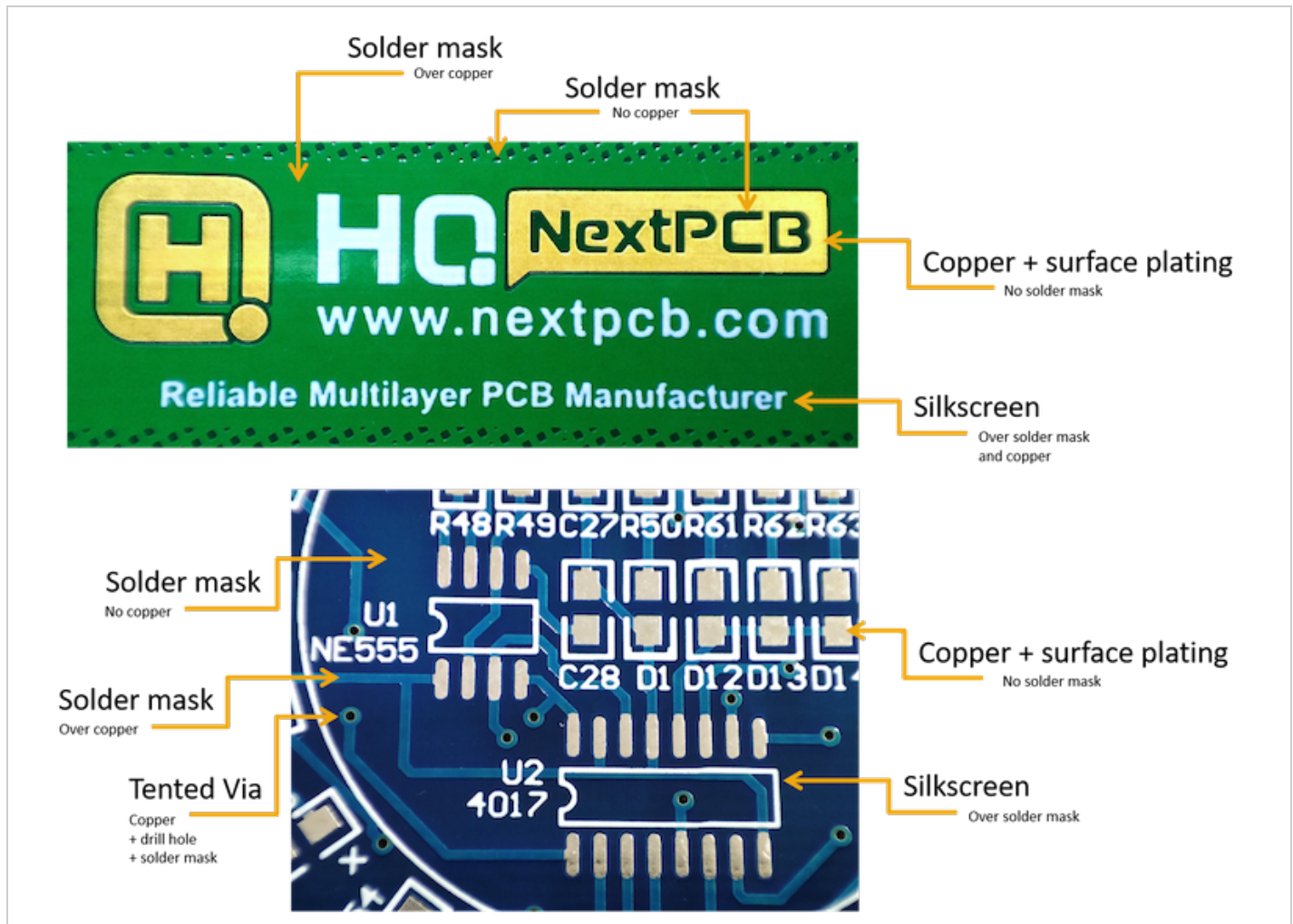


Figure 1. How Gerber layers combine to produce different features on a PCB board.

A Detailed Look at Gerber PCB Layers

In the following subsections, we will take a detailed look at these Gerber file layers:

- Top silkscreen
- Top solder mask
- Top copper
- Bottom copper
- Bottom solder mask
- Bottom silkscreen
- Outline / mechanical
- Drill
- Inner copper

Top Silkscreen (GTO)

Purpose: The top silkscreen Gerber file contains data to be printed, typically the white or black text on the PCB board, such as part designators, component outlines, logos, polarity indicators, etc.

Notes: This layer is optional. It is entirely OK to have a PCB without silkscreen if that is what you want.



Figure 2. Gerber top silkscreen layer.

Top Solder Mask (GTS)

Purpose: Contains data for the protective solder mask which protects the copper. This is typically a green film but can come in a variety of colors.

Notes: Solder mask layers are negative—meaning filled areas correspond to openings in the solder mask where no solder mask will be applied. These filled areas in the Gerber solder mask layer will leave copper exposed in the final product. This exposed copper is used for component attach and test access points.



Figure 3. Gerber top solder mask layer.

Top Copper Layer (GTL)

Purpose: Contains data for all areas requiring copper, including traces, pads, copper pour, vias, plated through holes, etc.

Notes: All filled areas correspond to areas with copper plating.



Figure 4. Gerber top copper mask layer.

Bottom Copper Layer (GBL)

Purpose: Identical to top copper, but flipped horizontally around the y-axis.

Notes: All filled areas correspond to areas with copper plating.

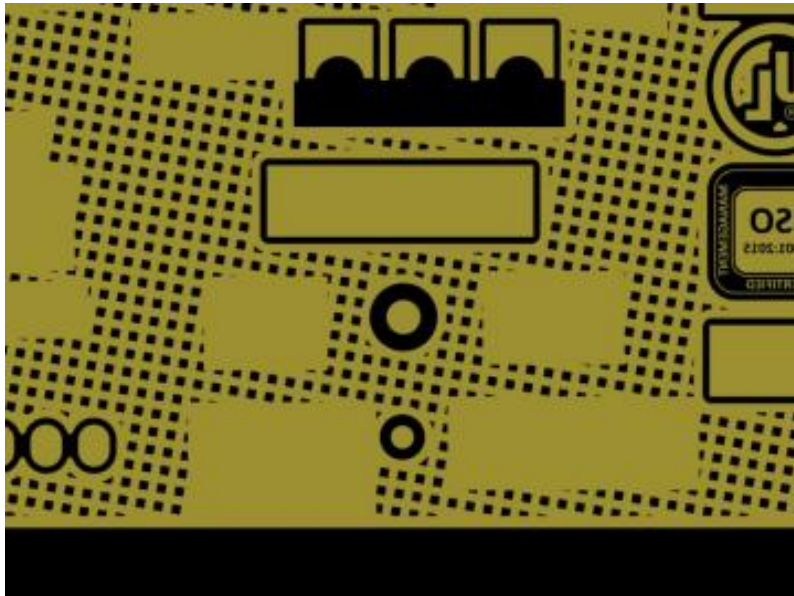


Figure 5. Gerber bottom copper mask layer.

Bottom Solder Mask (GBS)

Purpose: Identical to the top solder mask, but flipped horizontally around the y-axis.

Notes: Solder mask layers are negative—meaning filled areas correspond to openings in the solder mask where no solder mask will be applied. These filled areas in the Gerber solder mask layer will leave copper exposed in the final product. This exposed copper is used for component attach and test access points.